GENE CARLO FRANCO BELINARIO

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Research Interests

- **Astrophysics**, **Cosmology**, **Computational Physics**, **Physics Education**
- Gravitational Waves, Gravitational Lensing, Cluster Cosmology, Dark Energy
- Applications of Data Analysis, Machine Learning, and Neural Networks

Education

- **Master of Science in Physics**, Feb 2023 – Jul 2025
  National Institute of Physics, University of the Philippines Diliman
  Thesis: *Constraining cosmic gravity with gravitational-wave standard sirens*
  Supervisor: *Michael Francis Ian Vega II, Ph.D.*
  Co-mentors: *Reginald Christian Bernardo, Ph.D. & Reinabelle Reyes, Ph.D.*
- **Bachelor of Science in Applied Physics (Instrumentation)**, Aug 2018 – Feb 2023
  National Institute of Physics, University of the Philippines Diliman
  Honors: *Magna Cum Laude*, General Weighted Average: 1.2843 (1.00 highest – 5.00 lowest)
  Thesis: *Gravitational lensing signatures in gravitational-wave signals from compact binary coalescences*
  Supervisor: *Reinabelle Reyes, Ph.D.*

Research Experience

- **Independent Researcher**, August 2026 – Present

  Project: ***Cluster Cosmology using Cosmological Simulations***
  - Investigating the spatial clustering of galaxy clusters and their dependence on halo mass and redshift.
  - Perform likelihood-based cosmological inference on cluster observables such as cluster abundance, cluster clustering, and cluster-galaxy lensing using data from cosmological simulations (i.e. Magneticum, Uchuu Simulations).

- **Research Intern**, June 2026 – Aug 2026
  Academia Sinica Institute of Astronomy and Astrophysics

  Project: ***Finding Blue Clusters: When did Galaxy Clusters Stop Forming Stars?***
  - Identified galaxy cluster candidates in Uchuu Data Release 2 across various simulation snapshots.
  - Investigated the evolution of star-forming and quenched clusters and their dependence on redshift, cluster mass, and environments.

Research Experience (continued)

- **Graduate Student Researcher**, Jan 2025 – Jul 2025
  Gravity Group | Gravity
  National Institute of Physics, University of the Philippines Diliman

  Project: ***Cosmology with Gravitational Waves***
  - Simulated mock gravitational-wave (bright) standard siren catalogs using various cosmological models and LISA forecasts on supermassive black hole binary populations.
  - Performed cosmological parameter estimation with mock bright sirens independently and jointly with electromagnetic probes (e.g. Type Ia Supernovae, Baryon Acoustic Oscillations, Cosmic Chronometers).
  - This project served as the basis of my master's thesis and was presented as a contributed talk at the 42nd Samahang Pisika ng Pilipinas (Physics Society of the Philippines) Physics Conference and Annual Meeting.

  Project: ***Cosmo-Learn: Code for Learning Cosmology Using Different Methods and Mock Data***
  - Co-developed the open-source Python-based tool `cosmo-learn` [1] that simulates cosmological data and benchmark modern statistical and machine learning inference methods.
  - Implemented the simulation pipeline for generating bright siren events using various cosmological models and supermassive black hole binary populations.

- **Undergraduate Student Researcher**, Aug 2020 – Feb 2023
  Data and Computation Group | D&C
  National Institute of Physics, University of the Philippines Diliman

  Project: ***Gravitational Lensing of Gravitational Waves from Binary Black Hole Mergers***
  - Investigated the detectability of gravitationally lensed gravitational waves in the the LIGO Hanford and LIGO Livingston detectors and demonstrated measurement biases in inferring physical properties of the lens using lensed waveforms.
  - This project served as the basis of my undergraduate thesis and was presented as a contributed talk at the 41st Samahang Pisika ng Pilipinas (Physics Society of the Philippines) Physics Conference and Annual Meeting.

Teaching Experience

- **Instructor of Physics**, Sept 2023 – Jul 2025
  National Institute of Physics, University of the Philippines Diliman
  - Taught lecture and laboratory classes on elementary mechanics, electromagnetism, optics, thermodynamics, and computational physics to undergraduate students majoring in engineering and natural sciences.
  - Graded problem sets and examinations in fundamental physics, classical mechanics, electromagnetic theory, and quantum physics classes of undergraduate students majoring in physics and applied physics.
  - Developed curricula in elementary mechanics and electromagnetism laboratory classes.

- **Teaching Associate**, Feb 2023 – Aug 2023
  National Institute of Physics, University of the Philippines Diliman
  - Taught lecture and laboratory classes on elementary mechanics, electromagnetism, optics, and thermodynamics to undergraduate students majoring in engineering and natural sciences.

Skills

Programming	▸ Python, C++, Fortran, Command Line, Git, LaTeX, HTML
Toolkits	▸ Numpy, SciPy, matplotlib, Open MPI
Software	▸ Jupyter, VS Code, Mathematica
Language	▸ English, Filipino

Workshops and Invited Talks

- **AstroAI Network Summer School 2026**, 24-28 August 2026, *Invited Seminar* by the Academia Sinica Institute of Astronomy and Astrophysics (ASIAA) & Leung Center for Cosmology and Particle Astrophysics (LeCosPA) held at the National Taiwan University, Taiwan. *Project:* "Using Variational Autoencoder for Generating Synthetic Galaxy Spectra"
- **ASIAA Summer Student Program 2026**, 29 June - 21 August 2026, *Research Program* by the Academia Sinica Institute of Astronomy and Astrophysics (ASIAA) held at the National Taiwan University, Taiwan.
- **APCTP Workshop on Gravitational Waves 2025**, 20-25 April 2025, *Invited Seminar* by the Asia Pacific Center for Theoretical Physics (APCTP) held at the Institute of Physics, Academia Sinica, Taiwan.
- **1st Jagna Space Encounters - Python for the Stars: Unraveling Space Mysteries with Astronomical Data**, 15-18 January 2025 held at the Research Center for Theoretical Physics (RCTP), Central Visayan Institute Foundation (CVIF), Philippines. *Talk:* "Constraints on Cosmic Expansion History with Standard Sirens"
- **42nd Samahang Pisika ng Pilipinas Physics Conference and Annual Meeting**, 03-06 July 2024 held at the Batangas State University, Philippines. *Contributed Talk:* "Constraining cosmic dark energy with bright standard sirens"
- **Summer School on Cosmology**, 17-28 June 2024, *Invited Seminar* held at The Abdus Salam International Center for Theoretical Physics (ICTP), Trieste, Italy.
- **LISA Analysis Tools Workshop**, 15-18 April 2024, *Invited Online Seminar*
- **Frontiers in Theoretical and Experimental Physics**, 11 March 2024, *Invited Seminar* held at the National Institute of Physics, University of the Philippines Diliman, Metro Manila, Philippines.
- **41st Samahang Pisika ng Pilipinas Physics Conference and Annual Meeting**, 19-21 July 2023 held at the Del Carmen Convention Center and Bayay Numancia, Philippines. *Contributed Talk:* "Gravitational lensing signatures in gravitational waves from binary black hole mergers"
- **6th GW Open Data Workshop**, 15-17 May 2022, *Invited Online Seminar* by the Gravitational Wave Open Science Center (GWOSC)
- **5th GW Open Data Workshop**, 23-25 May 2022, *Invited Online Seminar* by the Gravitational Wave Open Science Center (GWOSC)

Research Publications

Journal Articles

1. R. C. Bernardo, D. Grandón, J. L. Said, V. H. Cárdenas, G. C. Belinario, and R. Reyes, "Cosmo-learn: Code for learning cosmology using different methods and mock data," *The European Physical Journal C*, 2026. DOI: https://doi.org/10.1140/epjc/s10052-026-15981-w.
2. E. D. Valentino, J. L. Said, and The CosmoVerse Network, "The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics," *Physics of the Dark Universe*, 2025. DOI: https://doi.org/10.1016/j.dark.2025.101965.

Conference Proceedings

1. G. C. Belinario, R. C. Bernardo, and R. Reyes, "Constraining cosmic dark energy using bright standard sirens," in *Proceedings of the Samahang Pisika ng Pilipinas*, vol. 42, 2024, SPP-2024-2E–01. URL: https://proceedings.spp-online.org/article/view/SPP-2024-2E-01.
2. J. Garrido, G. C. Belinario, J. Rodelas, and J. P. Esguerra, "Lessons from the First Philippine Junior Physics Challenge Examination," in *Proceedings of the Samahang Pisika ng Pilipinas*, vol. 42, 2024, SPP-2024-1E–01. URL: https://proceedings.spp-online.org/article/view/SPP-2024-1E-01.
3. G. C. Belinario and R. Reyes, "Gravitational lensing signatures in gravitational waves from binary black hole mergers," in *Proceedings of the Samahang Pisika ng Pilipinas*, vol. 41, 2023, SPP-2023-1G–04. URL: https://proceedings.spp-online.org/article/view/SPP-2023-1G-04.

References

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Reinabelle Reyes, Ph.D.
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Recognitions

- **Magna Cum Laude**, Jul 2023
  Bachelor of Science in Applied Physics (Instrumentation)
  College of Science, University of the Philippines Diliman
- **Merit Scholar**, Aug 2018 – Jul 2022
  Science Education Institute, Department of Science and Technology
  Republic of the Philippines
- **University Scholar**, Academic Year 2018 – 2023
  College of Science, University of the Philippines Diliman
- **College Scholar**, Academic Year 2018 – 2023
  College of Science, University of the Philippines Diliman

Involvements

Academic

- **Graduate Student Researcher**, Jan 2025 – Jul 2025
  Gravity Group, National Institute of Physics, University of the Philippines Diliman | Gravity
- **Graduate Student Researcher**, Feb 2023 – Dec 2024
  Data and Computation Group, National Institute of Physics, University of the Philippines Diliman | D&C
- **Undergraduate Student Researcher**, Aug 2020 – Feb 2023
  Data and Computation Group, National Institute of Physics, University of the Philippines Diliman | D&C

Administrative

- **Examination Committee**, Nov 2024 – Mar 2025
  Philippine Junior Physics Challenge 2025
- **Head of Technical Services Committee**, Sept 2024 – Jul 2025
  Junior Faculty, National Institute of Physics, University of the Philippines Diliman | NIP
- **Examination Committee**, Dec 2023 – Mar 2024
  Philippine Junior Physics Challenge 2024
- **Member-at-Large** & **Co-founder**, Mar 2021 – Sept 2022
  Students for the Exploration and Development of Space in the Philippines | SEDSPH
- **Vice President for Academic Affairs**, Sep 2020 – Jul 2021
  UP Physics Association, National Institute of Physics, University of the Philippines Diliman | UPPA

Community

- **Member**, Oct 2024 – Present
  Observational Cosmology and Systematics Working Group, CosmoVerse CA21136 | CosmoVerse COST Action
- **Member**, Oct 2024 – Present
  Data Analysis in Cosmology Working Group, CosmoVerse CA21136 | CosmoVerse COST Action

Miscellaneous

- **Referee**, Jul 2023 – Jul 2024
  Proceedings of the Samahang Pisika ng Pilipinas | SPP
- **Contributor**, Sept 2020
  Gravity Spy Citizen Science Project, Zooniverse | Gravity Spy